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Module 2: Network Discovery and Scanning with Nmap

Course: Introduction to Cybersecurity

Date: March 25, 2026

Environment: Kali Linux 2025.4 on VirtualBox

PRACTICAL WORK REPORT

1. Introduction

This report presents the results of the Nmap Network Awareness Lab Hands-on. The goal was to discover active hosts, spot open ports, identify running services, and estimate the operating system of the target machines.

The lab was carried out in a virtualized environment: Kali Linux 2025.4 installed on VirtualBox, configured in "Host-Only" network mode, with the IP address 192.168.56.102.

2. Work Environment

Component: Host System

Detail: Windows (physical PC)

Component: Virtual Machine

Detail: Kali Linux 2025.4

Component: Virtualization Tool

Detail: VirtualBox

Component: Network Mode

Detail: Host-Only

Component: Kali Machine IP

Detail: 192.168.56.102

Component: Nmap version

Detail: 7.95

Component: Scanned Network

Detail: 192.168.56.0/24

3. Part 1: Verification of the Installation

Command used

nmap --version

Result obtained



Nmap version 7.95

Platform: x86_64-pc-linux-gnu

Observation

Nmap is already installed on Kali Linux. Version 7.95 is functional and adapted to the manipulations requested.

4. Part 2: Discovering Active Hosts

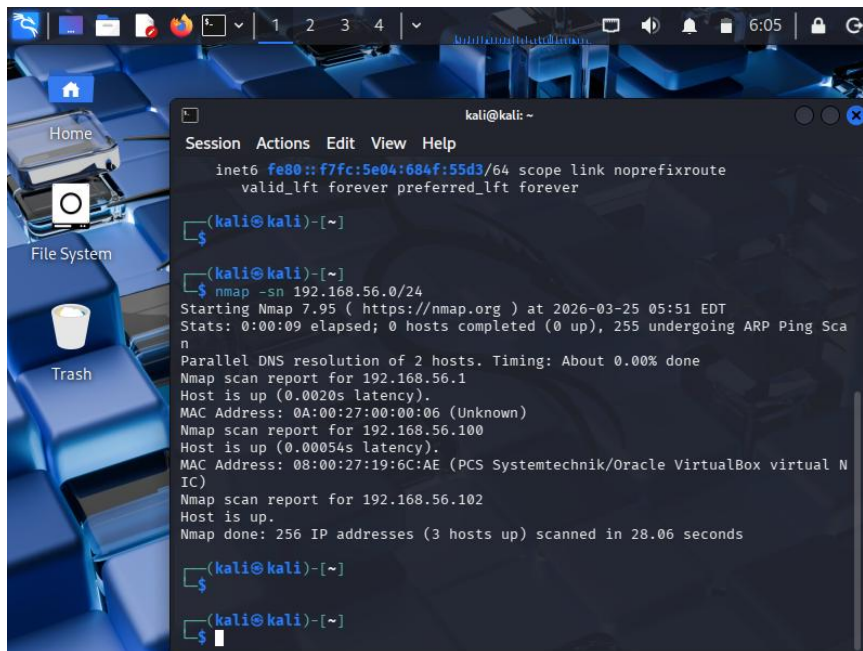
Command used

```
nmap -sn 192.168.56.0/24
```

Explanation

The `-sn` option performs a ping scan. It detects active machines without scanning their ports, scanning the entire 192.168.56.0/24 subnet.

Results achieved



IP Address: 192.168.56.1

Status: Active

Description: Windows Host PC (VirtualBox Host)

IP address: 192.168.56.100

Status: Active

Description: VirtualBox DHCP Server

IP address: 192.168.56.102

Status: Active

Description: Kali Linux Machine (our position)

Result: 3 active hosts detected in 28.06 seconds.

5. Part 3: Scanning Open Ports

Command used

nmap 192.168.56.1

Explanation

This command initiates a TCP scan of the 1000 most common ports on target 192.168.56.1 (host Windows PC).

Results achieved

```
kali@kali: ~  
Session Actions Edit View Help  
MAC Address: 0A:00:27:00:00:06 (Unknown)  
Nmap scan report for 192.168.56.100  
Host is up (0.00054s latency).  
MAC Address: 08:00:27:19:6C:AE (PCS Systemtechnik/Oracle VirtualBox virtual N  
IC)  
Nmap scan report for 192.168.56.102  
Host is up.  
Nmap done: 256 IP addresses (3 hosts up) scanned in 28.06 seconds  
  
(kali@kali)-[~]  
$  
  
(kali@kali)-[~]  
$ nmap 192.168.56.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-25 06:08 EDT  
Nmap scan report for 192.168.56.1  
Host is up (0.0014s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 0A:00:27:00:00:06 (Unknown)  
  
Nmap done: 1 IP address (1 host up) scanned in 18.22 seconds  
  
(kali@kali)-[~]  
$
```

Port: 135/tcpStatus: OpenService: msrpc

Port: 139/tcpStatus: OpenService: netbios-ssn

Port: 445/tcpStatus: OpenService: microsoft-ds

Summary: 997 ports filtered by the firewall. 3 open TCP ports.

6. Part 4: Service and Version Detection

Command used

```
nmap -sV 192.168.56.1
```

Explanation

The -sV option triggers version detection. Nmap sends probes to precisely identify exposed services.

Results achieved



Port: 135/tcp

Service: msrcpc

Version Detected: Microsoft Windows RPC

Port: 139/tcp

Service: netbios-ssn

Version Detected: Microsoft Windows netbios-ssn

Port: 445/tcp

Service: microsoft-ds

Version Detected: Not Precisely Identified

Additional information: OS = Windows (CPE: cpe:/o:microsoft:windows)

7. Part 5: Operating System Detection

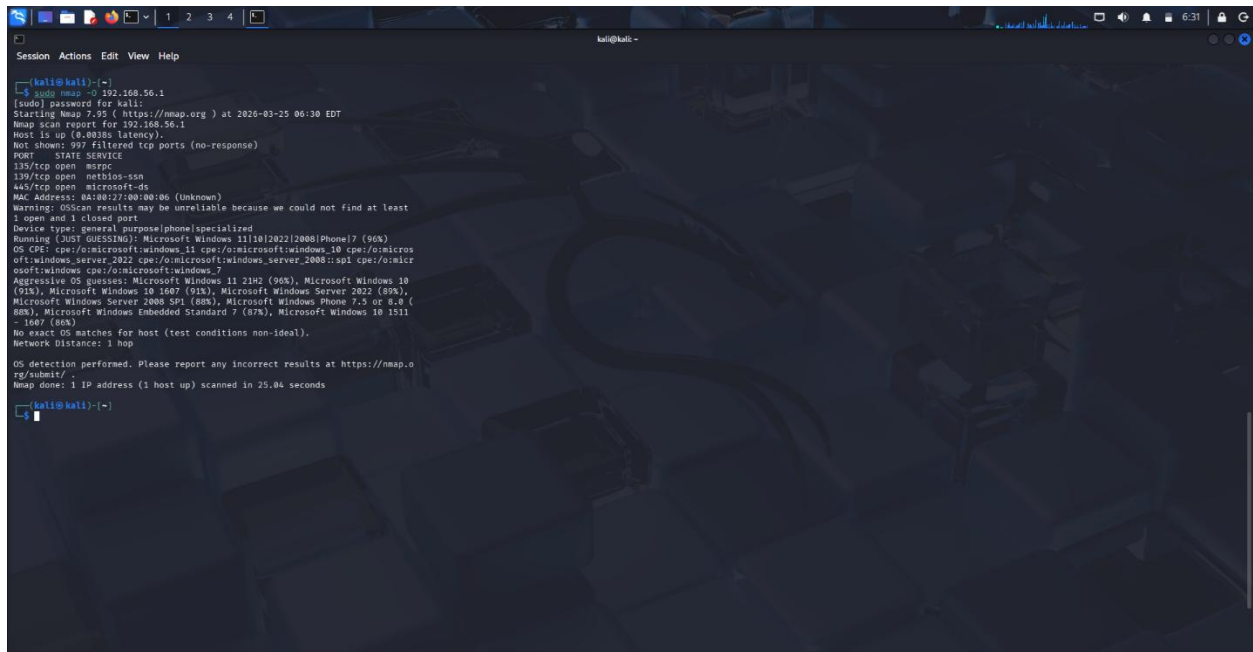
Command used


```
sudo nmap -o 192.168.56.1
```

Explanation

The -O option enables OS detection by comparing network responses to a database of known signatures.

Results achieved



```
kali@kali:~$ sudo nmap -O 192.168.56.1
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-25 06:30 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0035s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 84:AA:22:78:6A:06 (Unknown)
Warning: OSScan results may be unreliable because we could not find at least
3 open and 1 closed port
Device type: general purpose|phone|specialized
Running (JUST GUESSING): Microsoft Windows 1118|2022|2008|Phone|7 (96%)
OS CPE: cpe:/o:microsoft:windows.11 cpe:/o:microsoft:windows.10 cpe:/o:microsoft:windows.server.2022 cpe:/o:microsoft:windows.server.2008::sp1 cpe:/o:microsoft:windows.7
Aggressive OS guesses: Microsoft Windows 11 21H2 (96%), Microsoft Windows 10 (91%), Microsoft Windows 10 1607 (91%), Microsoft Windows Server 2022 (89%), Microsoft Windows Server 2008 SP1 (88%), Microsoft Windows Phone 7.5 or 8.0 (88%), Microsoft Windows Embedded Standard 7 (87%), Microsoft Windows 10 1511 - 1607 (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
OS detection performed. Please report any incorrect results at https://nmap.org
nmap done: 1 IP address (1 host up) scanned in 25.04 seconds
kali@kali:~$
```

OS Detected: Microsoft Windows 11 21H2Probability: 96%

OS Detected: Microsoft Windows 10Probability: 91%

OS Detected: Microsoft Windows Server 2022Probability: 89%

OS Detected: Microsoft Windows Server 2008 SP1Probability: 88%

Most likely system: Windows 11 21H2 (96%)Note: The message "OSScan results may be unreliable" is expected when the firewall filters ports.

8. Part 6: Aggressive Scanning

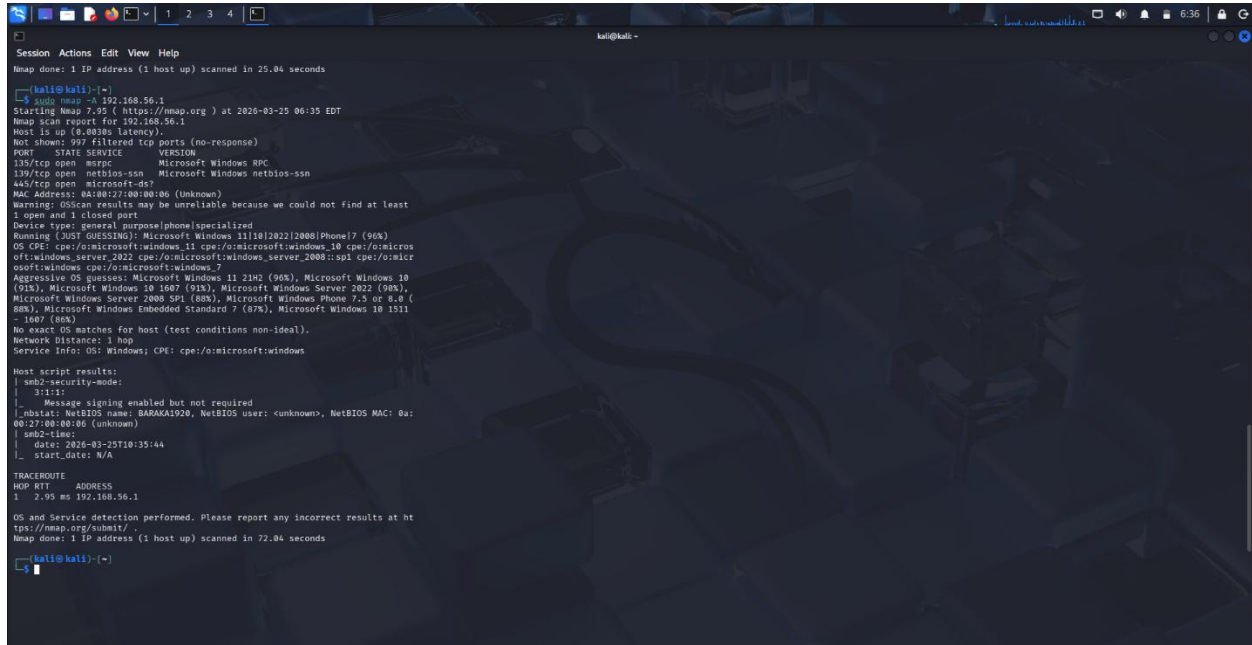
Command used

```
sudo nmap -A 192.168.56.1
```

Explanation

-A mode combines OS detection, version detection, NSE scripts, and a route trace. This is the most complete scan.

Additional information obtained



```
Session Actions Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 25.04 seconds

kali@kali:~$ sudo nmap -A 192.168.56.1
Starting Nmap 7.92 ( https://nmap.org ) at 2020-03-25 06:35 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0030s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
MAC Address: 0A:80:27:00:00:00 (unknown)
Warning: OSscan results may be unreliable because we could not find at least
1 open and 1 closed port
Device type: general purpose|phone|specialized
Running (3251 GUESSES): Microsoft Windows 1111a|2022|2000|Phone|7 (96%)
OS CPE: cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_10 cpe:/o:micros
oft:windows_server_2022 cpe:/o:microsoft:windows_server_2008::sp1 cpe:/o:micr
osoft:windows_cpe:/o:microsoft:windows_7
Aggressive OS guesses: Microsoft Windows 11 21H2 (96%), Microsoft Windows 10
(91%), Microsoft Windows 10 1607 (91%), Microsoft Windows Server 2022 (90%),
Microsoft Windows Server 2008 SP1 (88%), Microsoft Windows Phone 7.5 or 8.0 (
88%), Microsoft Windows Embedded Standard 7 (87%), Microsoft Windows 10 1511
- 1607 (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb2-security-mode:
|   3:111:
|     Message signing enabled but not required
|_ smb2-time:
|   date: 2026-03-25T10:35:44
|_ start_date: N/A

TRACEROUTE
HOP RTT ADDRESS
1 2.95 ms 192.168.56.1

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 72.04 seconds

kali@kali:~$
```

NetBIOS name: BARAKA1920

SMB signing enabled but not required (potentially risky configuration)

Synchronized date and time: 2026-03-25T10:35:44

Network distance: 1 hop

Confirmed OS: Windows 11 21H2 (96%)

9. Answers to Analysis Questions

Question 1: How many hosts have been detected?

Three active hosts:

192.168.56.1: Windows Host PC

192.168.56.100: VirtualBox DHCP Server

192.168.56.102: Kali Linux (our machine)

Question 2: Which ports are the most common?

Frequently opened ports on Windows:

- 135 (MSRPC): Remote Procedure Calls
- 139 (NetBIOS): File/Printer Sharing
- 445 (SMB): Microsoft Network Share

These ports are typical of a Windows workstation connected to the network.

Question 3: Why are open ports a risk?

An open port means a service that can be accessed remotely.

If the service is vulnerable, it can be exploited.

Port 445 (SMB) was used by WannaCry (2017).

Port 135 (RPC) has been targeted by worms such as Blaster (2003).

More open ports = larger attack surface.

Question 4: What information can an attacker collect?

Active IP addresses

Open ports and available services

Exact versions of services

Probable operating system

Machine Name (NetBIOS)

Network topology (hops, routes)

This data corresponds to the reconnaissance phase of an attack.

10. Safety Observations

Windows Services Exposure

Ports 135, 139, and 445 are required over LAN but should never be exposed to the Internet unprotected.

Importance of the Firewall

The firewall has filtered 997 ports out of 1000, which greatly reduces the attack surface.

SMB Risk

The message "SMB signing enabled but not required" indicates a risky configuration. The lack of a mandatory signature facilitates NTLM relay attacks.

Recommendations

Disable NetBIOS if unnecessary

Enable mandatory SMB signing

Update the OS regularly

Deploy an IDS to detect scans

Periodically audit open ports

11. Conclusion

This lab allowed them to practice the basics of network reconnaissance:

- Active Host Detection
- Identifying Ports and Services
- Operating System Estimation

Recognition is a key phase in cybersecurity. The better an attacker knows the topology and services, the more effectively they can target. Reducing the attack surface, continuously updating and monitoring the network remains a must.